

CHAPTER 6

6. RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

Definition: A random variable is a numerical description of the outcomes of the experiment or a numerical valued function defined on sample space, usually denoted by capital letters.

Example: If X is a random variable, then it is a function from the elements of the sample space to the set of real numbers. i.e.

X is a function $X: S \rightarrow R$

→ A random variable takes a possible outcome and assigns a number to it.

Example: Flip a coin three times, let X be the number of heads in three tosses.

$\Rightarrow S = \{(HHH), (HHT), (HTH), (HTT), (THH), (THT), (TTH), (TTT)\}$

$\Rightarrow X(HHH) = 3,$

$X(HHT) = X(HTH) = X(THH) = 2,$

$X(HTT) = X(THT) = X(TTH) = 1$

$X(TTT) = 0$

$X = \{0, 1, 2, 3\}$

→ X assumes a specific number of values with some probabilities.

Random variables are of two types:

1. Discrete random variable: are variables which can assume only a specific number of values. They have values that can be counted

Examples:

- Toss coin n times and count the number of heads.
- Number of children in a family.
- Number of car accidents per week.
- Number of defective items in a given company.
- Number of bacteria per two cubic centimeter of water.

2. Continuous random variable: are variables that can assume all values between any two give values.

Examples:

- Height of students at certain college.
- Mark of a student.
- Life time of light bulbs.
- Length of time required to complete a given training.

Definition: a probability distribution consists of a value a random variable can assume and the corresponding probabilities of the values.

Example: Consider the experiment of tossing a coin three times. Let X be the number of heads. Construct the probability distribution of X .

Solution:

- First identify the possible value that X can assume.
- Calculate the probability of each possible distinct value of X and express X in the form of frequency distribution.

$X = x$	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

Probability distribution is denoted by P for discrete and by f for continuous random variable.

Properties of Probability Distribution:

1.

$$P(x) \geq 0, \text{ if } X \text{ is discrete.}$$

$$f(x) \geq 0, \text{ if } X \text{ is continuous.}$$

2.

$$\sum_x P(X = x) = 1, \text{ if } X \text{ is discrete.}$$

$$\int_x f(x) dx = 1, \text{ if } X \text{ is continuous.}$$

Note:

1. If X is a continuous random variable then

$$P(a < X < b) = \int_a^b f(x)dx$$

2. Probability of a fixed value of a continuous random variable is zero.

$$\Rightarrow P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

3. If X is discrete random variable the

$$P(a < X < b) = \sum_{x=a+1}^{b-1} P(x)$$

$$P(a \leq X < b) = \sum_{x=a}^{b-1} p(x)$$

$$P(a < X \leq b) = \sum_{x=a+1}^b P(x)$$

$$P(a \leq X \leq b) = \sum_{x=a}^b P(x)$$

4. Probability means area for continuous random variable.

Introduction to expectation

Definition:

1. Let a discrete random variable X assume the values $X_1, X_2, \dots, X_n$ with the probabilities $P(X_1), P(X_2), \dots, P(X_n)$ respectively. Then the expected value of X , denoted as $E(X)$ is defined as:

$$\begin{aligned} E(X) &= X_1P(X_1) + X_2P(X_2) + \dots + X_nP(X_n) \\ &= \sum_{i=1}^n X_iP(X_i) \end{aligned}$$

2. Let X be a continuous random variable assuming the values in the

interval (a, b) such that $\int_a^b f(x)dx = 1$, then

$$E(X) = \int_a^b x f(x)dx$$

Examples:

1. What is the expected value of a random variable X obtained by tossing a coin three times where X is the number of heads

Solution:

First construct the probability distribution of X

$X = x$	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

$$\begin{aligned}\Rightarrow E(X) &= X_1P(X_1) + X_2P(X_2) + \dots + X_nP(X_n) \\ &= 0 \cdot 1/8 + 1 \cdot 3/8 + \dots + 2 \cdot 1/8 \\ &= 1.5\end{aligned}$$

2. Suppose a charity organization is mailing printed return-address stickers to over one million homes in the Ethiopia. Each recipient is asked to donate either \$1, \$2, \$5, \$10, \$15, or \$20. Based on past experience, the amount a person donates is believed to follow the following probability distribution:

$X = x$	\$1	\$2	\$5	\$10	\$15	\$20
$P(X = x)$	0.1	0.2	0.3	0.2	0.15	0.05

What is expected that an average donor to contribute?

Solution:

$X = x$	\$1	\$2	\$5	\$10	\$15	\$20	Total
$P(X = x)$	0.1	0.2	0.3	0.2	0.15	0.05	1
$xP(X = x)$	0.1	0.4	1.5	2	2.25	1	7.25

$$\Rightarrow E(X) = \sum_{i=1}^6 x_i P(X = x_i) = \$7.25$$

Mean and Variance of a random variable

Let X be given random variable.

1. The expected value of X is its mean

$$\Rightarrow \text{Mean of } X = E(X)$$

2. The variance of X is given by:

$$\text{Variance of } X = \text{var}(X) = E(X^2) - [E(X)]^2$$

Where:

$$E(X^2) = \sum_{i=1}^n x_i^2 P(X = x_i) \text{ , if } X \text{ is discrete}$$

$$= \int x^2 f(x) dx \text{ , if } X \text{ is continuous.}$$

Examples:

1. Find the mean and the variance of a random variable X in example 2 above.

Solutions:

$X = x$	\$1	\$2	\$5	\$10	\$15	\$20	Total
$P(X = x)$	0.1	0.2	0.3	0.2	0.15	0.05	1
$xP(X = x)$	0.1	0.4	1.5	2	2.25	1	7.25
$x^2P(X = x)$	0.1	0.8	7.5	20	33.75	20	82.15

$$\Rightarrow E(X) = 7.25$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 82.15 - 7.25^2 = 29.59$$

2. Two dice are rolled. Let X be a random variable denoting the sum of the numbers on the two dice.

- i) Give the probability distribution of X
- ii) Compute the expected value of X and its variance

Solution (exercise)

There are some general rules for mathematical expectation.

Let X and Y are random variables and k be a constant.

RULE 1

$$E(k) = k$$

RULE 2

$$Var(k) = 0$$

RULE 3

$$E(kX) = kE(X)$$

RULE 4

$$Var(kX) = k^2 Var(X)$$

RULE 5

$$E(X + Y) = E(X) + E(Y)$$

Common Discrete Probability Distributions**1. Binomial Distribution**

A binomial experiment is a probability experiment that satisfies the following four requirements called assumptions of a binomial distribution.

1. The experiment consists of n identical trials.
2. Each trial has only one of the two possible mutually exclusive outcomes, success or a failure.
3. The probability of each outcome does not change from trial to trial, and
4. The trials are independent, thus we must sample with replacement.

Examples of binomial experiments

- Tossing a coin 20 times to see how many tails occur.
- Asking 200 people if they watch BBC news.
- Registering a newly produced product as defective or non defective.
- Asking 100 people if they favour the ruling party.

- Rolling a die to see if a 5 appears.

Definition: The outcomes of the binomial experiment and the corresponding probabilities of these outcomes are called **Binomial Distribution**.

Let P = the probability of success

$q = 1 - p$ = the probability of failure on any given trial

Then the probability of getting x successes in n trials becomes:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

And this is some times written as:

$$X \sim \text{Bin}(n, p)$$

When using the binomial formula to solve problems, we have to identify three things:

- The number of trials (n)
- The probability of a success on any one trial (p) and
- The number of successes desired (X).

Examples:

1. What is the probability of getting three heads by tossing a fair coin four times?

Solution:

Let X be the number of heads in tossing a fair coin four times

$$X \sim \text{Bin}(n = 4, p = 0.50)$$

$$\begin{aligned} \Rightarrow P(X = x) &= \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, 3, 4 \\ &= \binom{4}{x} 0.5^x 0.5^{4-x} = \binom{4}{x} 0.5^4 \Rightarrow P(X = 3) = \binom{4}{3} 0.5^4 = 0.25 \end{aligned}$$

2. Suppose that an examination consists of six true and false questions, and assume that a student has no knowledge of the subject matter. The probability that the student will guess the correct answer to the first question is 30%. Likewise, the probability of guessing each of the remaining questions correctly is also 30%.
- What is the probability of getting more than three correct answers?
 - What is the probability of getting at least two correct answers?
 - What is the probability of getting at most three correct answers?
 - What is the probability of getting less than five correct answers?

Solution

Let X = the number of correct answers that the student gets.

$$X \sim \text{Bin}(n = 6, p = 0.30)$$

$$\text{a) } P(X > 3) = ?$$

$$\begin{aligned} \Rightarrow P(X = x) &= \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, 6 \\ &= \binom{6}{x} 0.3^x 0.7^{6-x} \end{aligned}$$

$$\begin{aligned} \Rightarrow P(X > 3) &= P(X = 4) + P(X = 5) + P(X = 6) \\ &= 0.060 + 0.010 + 0.001 \\ &= 0.071 \end{aligned}$$

Thus, we may conclude that if 30% of the exam questions are answered by guessing, the probability is 0.071 (or 7.1%) that more than four of the questions are answered correctly by the student.

$$\text{b) } P(X \geq 2) = ?$$

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) \\ &= 0.324 + 0.185 + 0.060 + 0.010 + 0.001 = 0.58 \end{aligned}$$

$$\text{c) } P(X \leq 3) = ?$$

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ = 0.118 + 0.303 + 0.324 + 0.18 = 0.93$$

$$d) P(X < 5) = ?$$

$$P(X < 5) = 1 - P(X \geq 5) \\ = 1 - \{P(X = 5) + P(X = 6)\} \\ = 1 - (0.010 + 0.001) = 0.989$$

Exercises:

1. Suppose that 4% of all TVs made by A&B Company in 2000 are defective. If eight of these TVs are randomly selected from across the country and tested, what is the probability that *exactly* three of them are defective? Assume that each TV is made independently of the others.
2. An allergist claims that 45% of the patients she tests are allergic to some type of weed. What is the probability that
 - a) Exactly 3 of her next 4 patients are allergic to weeds?
 - b) None of her next 4 patients are allergic to weeds?
3. Explain why the following experiments are not Binomial
 - Rolling a die until a 6 appears.
 - Asking 20 people how old they are.
 - Drawing 5 cards from a deck for a poker hand.

Remark: If X is a binomial random variable with parameters n and p then

$E(X) = np, \quad Var(X) = npq$

2. Poisson Distribution

- A random variable X is said to have a Poisson distribution if its probability distribution is given by:

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

Where λ = the average number.

- The Poisson distribution depends *only* on the average number of occurrences per unit time of space.
- The Poisson distribution is used as a distribution of rare events, such as:
 - Number of misprints.
 - Natural disasters like earth quake.
 - Accidents.
 - Hereditary.
 - Arrivals
- The process that gives rise to such events are called Poisson process.

Examples:

1. If 1.6 accidents can be expected an intersection on any given day, what is the probability that there will be 3 accidents on any given day?

Solution; Let X = the number of accidents, $\lambda = 1.6$

$$X = \text{poisson}(1.6) \Rightarrow p(X = x) = \frac{1.6^x e^{-1.6}}{x!}$$

$$p(X = 3) = \frac{1.6^3 e^{-1.6}}{3!} = 0.1380$$

2. On the average, five smokers pass a certain street corners every ten minutes, what is the probability that during a given 10 minutes the number of smokers passing will be
 - a. 6 or fewer
 - b. 7 or more
 - c. Exactly 8..... (Exercise)

If X is a Poisson random variable with parameters λ then

$E(X) = \lambda, \quad Var(X) = \lambda$
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Note:

The Poisson probability distribution provides a close approximation to the binomial probability distribution when n is large and p is quite small or quite large with $\lambda = np$.

$$P(X = x) = \frac{(np)^x e^{-(np)}}{x!}, \quad x = 0, 1, 2, \dots$$

Where $\lambda = np = \text{the average number}$.

Usually we use this approximation if $np \leq 5$. In other words, if $n > 20$ and $np \leq 5$ [or $n(1-p) \leq 5$], then we may use Poisson distribution as an approximation to binomial distribution.

Example:

- Find the binomial probability $P(X=3)$ by using the Poisson distribution if $p = 0.01$ and $n = 200$

Solution:

Using Poisson, $\lambda = np = 0.01 * 200 = 2 \Rightarrow P(X = 3) = \frac{2^3 e^{-2}}{3!} = 0.1804$

Using Binomial, $n = 200, p = 0.01$

$$\Rightarrow P(X = 3) = \binom{200}{3} (0.01)^3 (0.99)^{197} = 0.1814$$

Common Continuous Probability Distributions**1. Normal Distribution**

A random variable X is said to have a normal distribution if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

Where $\mu = E(X)$, $\sigma^2 = \text{Variance}(X)$

μ and σ^2 are the Parameters of the Normal Distribution.

Properties of Normal Distribution:

1. It is bell shaped and is symmetrical about its mean and it is mesokurtic.

The maximum ordinate is at $x = \mu$ and is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}$$

2. It is asymptotic to the axis, i.e., it extends indefinitely in either direction from the mean.
3. It is a continuous distribution.
4. It is a family of curves, i.e., every unique pair of mean and standard deviation defines a different normal distribution. Thus, the normal distribution is completely described by two parameters: mean and standard deviation.
5. Total area under the curve sums to 1, i.e., the area of the distribution on each side of the mean is 0.5. $\Rightarrow \int_{-\infty}^{\infty} f(x)dx = 1$
6. It is unimodal, i.e., values mound up only in the center of the curve.
7. *Mean = Median = mode* = μ
8. The probability that a random variable will have a value between any two points is equal to the area under the curve between those points.

Note: To facilitate the use of normal distribution, the following distribution known as the standard normal distribution was derived by using the transformation

$$Z = \frac{X - \mu}{\sigma} \Rightarrow f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

Properties of the Standard Normal Distribution:

Same as a normal distribution, but also...

- Mean is zero
- Variance is one

- Standard Deviation is one
- Areas under the standard normal distribution curve have been tabulated in various ways. The most common ones are the areas between

$Z = 0$ and a positive value of Z .

- Given a normal distributed random variable X with

Mean μ and standard deviation σ

$$P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right)$$

$$\Rightarrow P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$$

Note:

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

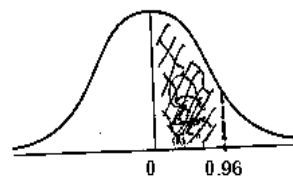
Examples:

1. Find the area under the standard normal distribution which lies

- a) Between $Z = 0$ and $Z = 0.96$

Solution:

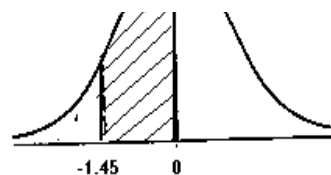
$$\text{Area} = P(0 < Z < 0.96) = 0.3315$$



- b) Between $Z = -1.45$ and $Z = 0$

Solution:

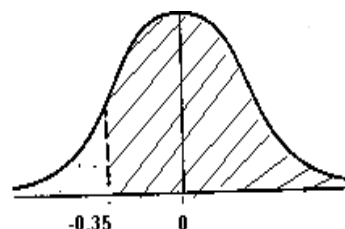
$$\begin{aligned} \text{Area} &= P(-1.45 < Z < 0) \\ &= P(0 < Z < 1.45) \\ &= 0.4265 \end{aligned}$$



- c) To the right of $Z = -0.35$

Solution:

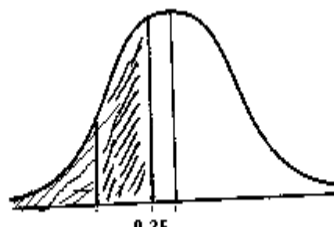
$$\begin{aligned} \text{Area} &= P(Z > -0.35) \\ &= P(-0.35 < Z < 0) + P(Z > 0) \\ &= P(0 < Z < 0.35) + P(Z > 0) \\ &= 0.1368 + 0.50 = 0.6368 \end{aligned}$$



- d) To the left of $Z = -0.35$

Solution:

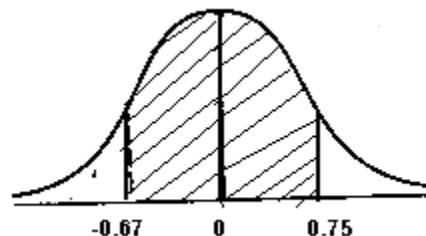
$$\begin{aligned} \text{Area} &= P(Z < -0.35) \\ &= 1 - P(Z > -0.35) \\ &= 1 - 0.6368 = 0.3632 \end{aligned}$$



e) Between $Z = -0.67$ and $Z = 0.75$

Solution:

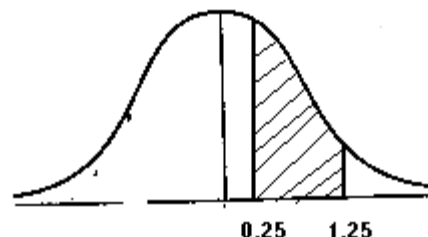
$$\begin{aligned} \text{Area} &= P(-0.67 < Z < 0.75) \\ &= P(-0.67 < Z < 0) + P(0 < Z < 0.75) \\ &= P(0 < Z < 0.67) + P(0 < Z < 0.75) \\ &= 0.2486 + 0.2734 = 0.5220 \end{aligned}$$



f) Between $Z = 0.25$ and $Z = 1.25$

Solution:

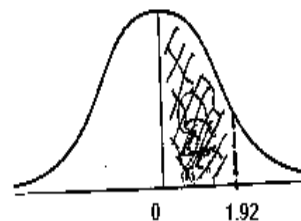
$$\begin{aligned} \text{Area} &= P(0.25 < Z < 1.25) \\ &= P(0 < Z < 1.25) - P(0 < Z < 0.25) = 0.3934 - 0.0987 = 0.2947 \end{aligned}$$



2. Find the value of Z if

a) The normal curve area between 0 and z(positive) is 0.4726

Solution



$$\begin{aligned} P(0 < Z < z) &= 0.4726 \text{ and from table} \\ P(0 < Z < 1.92) &= 0.4726 \Leftrightarrow z = 1.92 \dots \text{uniqueness of Area.} \end{aligned}$$

b) The area to the left of z is 0.9868

Solution

$$P(Z < z) = 0.9868$$

$$\begin{aligned} &= P(Z < 0) + P(0 < Z < z) = 0.50 + P(0 < Z < z) \Rightarrow P(0 < Z < z) = \\ &0.9868 - 0.50 = 0.4868 \text{ and from table } P(0 < Z < 2.2) = 0.4868 \Leftrightarrow z = 2.2 \end{aligned}$$

3. A random variable X has a normal distribution with mean 80 and standard deviation 4.8. What is the probability that it will take a value
- a) Less than 87.2
 - b) Greater than 76.4
 - c) Between 81.2 and 86.0

Solution

X is normal with mean, $\mu = 80$, standard deviation, $\sigma = 4.8$

a)

$$P(X < 87.2) = P\left(\frac{X - \mu}{\sigma} < \frac{87.2 - \mu}{\sigma}\right) = P\left(Z < \frac{87.2 - 80}{4.8}\right) = P(Z < 1.5) = P(Z < 0) + P(0 < Z < 1.5) = 0.50 + 0.4332 = \underline{\underline{0.9332}}$$

b)

$$P(X > 76.4) = P\left(\frac{X - \mu}{\sigma} > \frac{76.4 - \mu}{\sigma}\right) = P\left(Z > \frac{76.4 - 80}{4.8}\right) = P(Z > -0.75) = P(Z > 0) + P(0 < Z < 0.75) = 0.50 + 0.2734 = \underline{\underline{0.7734}}$$

c)

4. A normal distribution has mean 62.4. Find its standard deviation if 20.0% of the area under the normal curve lies to the right of 72.9

Solution

$$\begin{aligned} P(X > 72.9) &= 0.2005 \Rightarrow P\left(\frac{X - \mu}{\sigma} > \frac{72.9 - \mu}{\sigma}\right) = 0.2005 \\ &\Rightarrow P\left(Z > \frac{72.9 - 62.4}{\sigma}\right) = 0.2005 \Rightarrow P\left(Z > \frac{10.5}{\sigma}\right) = 0.2005 \\ &\Rightarrow P\left(0 < Z < \frac{10.5}{\sigma}\right) = 0.50 - 0.2005 = 0.2995 \\ &\text{And from table } P(0 < Z < 0.84) = 0.2995 \\ &\Leftrightarrow \frac{10.5}{\sigma} = 0.84 \Rightarrow \sigma = \underline{\underline{12.5}} \end{aligned}$$

4. A random variable has a normal distribution with $\sigma = 5$. Find its mean if the probability that the random variable will assume a value less than 52.5 is 0.6915.

Solution

$$P(Z < z) = P\left(Z < \frac{52.5 - \mu}{5}\right) = 0.6915$$

$$\Rightarrow P(0 < Z < z) = 0.6915 - 0.50 = 0.1915.$$

$$\text{But from the table} \Rightarrow P(0 < Z < 0.5) = 0.1915$$

$$\Leftrightarrow z = \frac{52.5 - \mu}{5} = 0.5$$

$$\Rightarrow \mu = \underline{\underline{50}}$$

- e) Of a large group of men, 5% are less than 60 inches in height and 40% are between 60 & 65 inches. Assuming a normal distribution, find the mean and standard deviation of heights.

Solution (Exercise)